

CHAPTER TWENTY

THE OCCULT SCIENCES IN SAFAVID IRAN AND SAFAVID OCCULT SCIENTISTS ABROAD



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POST-MONGOL ISLAMICATE IMPERIALISM: THE CONTEST FOR SACRAL POWER

Jean Chardin, Huguenot sojourner in Iran under Shāh Soleyman (r. 1666–94), patronizingly observed of Safavid elite society: ‘Most [Persians] are dependent on astrologers and other diviners as a child is dependent on its nurse’.¹ Modern historians are no less patronizing with respect to the enduring Safavid preoccupation with occultism; but rather than deride it, like Chardin, most discreetly ignore the phenomenon altogether. The overwhelming majority of studies on the period mention occult sciences (*‘olum ghariba*)² not at all or only in merest passing, while those intellectual histories of Safavid philosophy-science that do acknowledge their popularity typically banish them, as an undifferentiated mass, to the dumping ground that is ‘mysticism’, or worse, ‘popular religion’, even ‘superstition’.

As recent, less occultophobic research is showing, however, occultism – i.e., an investment in one or more of the occult sciences – was a mainstream pursuit of scholarly and ruling elites throughout the post-Mongol Persianate world, including the Ottoman, Safavid and Mughal Empires; indeed, this is one of the features that most distinguishes early modern Islamicate societies from their antecedents of the so-called classical period, and even more so from the successor states of the colonial period. In the Persianate world, the formative fourteenth–sixteenth centuries witnessed an occultist arms race, as it were, for messianic and sacral forms of political legitimacy, which reached a crescendo with the approach of the Islamic millennium (1592 CE). As in any arms race, ruling dynasts eagerly cultivated professional specialists in the relevant field – to wit, the art and science of *valāya*, ‘sacral power’. The Mongol destruction of the reigning, but brittle, caliphal-sultanic-jurisprudential model of Islamicate societies in the mid-thirteenth century had created a vacuum of legitimacy into which rushed this Shi‘i-Sufi quicksilver category, far more flexible and adaptable than the model it replaced, and it soon became hegemonic throughout the Persianate world. Securing access to this sacral power became a driving concern

of ruling and scholarly elites, whether by way of Sufism, occultism or Alidism, and often eclectic combinations of all three.

The Safavids, to at least the same degree as their Ottoman, Mughal and Uzbek rivals, opted for this combinative strategy during the sixteenth century. The militarized Shi'i-Sufi messianic fervor of Shāh Esmā'il I (r. 1501–24) – self-proclaimed reincarnation of 'Ali and incarnation of God – and his fanatical Turkmen Qezelbāsh horde was indeed responsible for creating a conquest state in Iran; but as a political framework it was too volatile to endure. Accordingly, Esmā'il's successor, Shāh Tahmāsb I (r. 1524–76), sought to routinize his father's messianic charisma by turning to orthodox Shi'ism—and by throwing himself into the study of the occult sciences, especially geomancy ('*elm al-raml*'). The Venetian diplomat Vincenzo degli Alessandri (d. after 1595) reported, for instance, that the shah had not left his palace for a decade, so devoted he was to practicing the science as substantiation for his claim to prophetlike sanctity and vision. (D'Alessandri further observed, incredulously, that Tahmāsb was openly worshiped by the Safavid populace as a godlike being possessed of a salvifically talismanic body.)³ This close identification of occultism with both Sufism and Shi'ism remained in effect through at least the turn of the seventeenth century, as contemporary scholarship attests. Most telling in this regard is the schema offered by Abo'l-Qāsem Ansāri Kāzeruni (fl. 1605), a prominent Shirazi scholar in the service of Shāh 'Abbās the Great (r. 1587–1629), in his *Ladder to the Heavens* (*Sollam al-samavāt*), an eclectic work devoted to constructing an intellectual-religious pedigree for his patron that is simultaneously Twelver Shi'i, Sufi and occultist. To this end, the *Sollam* includes a long chapter taxonomizing the occult sciences as a subset not of natural or mathematical philosophy, as they were usually classified from Farābi and Ebn Sinā onward, but exclusively of *valaya*, here presented as Sufi-style sainthood rather than a Twelver theological category.⁴ Similarly, the *Gardens of the Righteous* (*Reyāz al-abrār*) of Hoseyn 'Āqeli Rostamdāri, a comprehensive Persian encyclopedia written in Qazvin in 1571 for Tahmāsb and treating of some ninety sciences in systematic fashion, features both a strong Twelver flavor and a heavy emphasis on the occult sciences. Most significantly, therein Rostamdāri too breaks with precedent to reclassify two occult sciences – astrology ('*elm al-nojum*') and geomancy – as mathematical, while strongly implying that most of the other occult sciences transcend even that category in their connection to the Imams as sole vectors of *valaya*, including in the first place lettrism ('*elm al-horuf*') and alchemy ('*elm al-kimiya*').⁵ Thus even the early Safavids, who embraced Shi'ism precisely as a means of cornering the religiopolitical market on Alid *valāya*, still considered occultism to be central to their imperial and theological claims, and among the most effective means of routinizing and harnessing the messianic-Sufi charisma of Esmā'il as divine conqueror.

In this, they were following the Timurid model to the letter. Amir Temūr's (r. 1370–1405) own irresistible charisma as world conqueror was encapsulated in the title Lord of Conjunction (*sāheb-qerān*), a purely astrological designation that entered regular use from the thirteenth century onward, then was definitively 'Timuridized' by historians working for Temūr's immediate successors.⁶ I have shown elsewhere that the same scholar-occultists who made this title so central to Timurid universalist imperial ideology relied on lettrism in equal measure, producing a highly distinctive

dual astrological-lettrist platform that was to remain popular among successor states for centuries as a primary means of signaling their Timurid-style universalist legitimacy.⁷

To understand the full significance of this ideological platform, it must be borne in mind that lettrism, by the early fifteenth century, was widely considered to be not only the most Islamic of all the occult sciences, but the most Islamic, and the most universal, of *all* sciences in its status as both the chiefest branch of the ‘science of divine unity’ (*‘elm al-towhid*) and the ‘science of the saints’ (*‘elm al-owliyā*) *par excellence*. So argued the ‘Greatest Master’ Ebn ‘Arabi (d. 1240), foremost theoretician of *valāya*, and his argument was first formalized in Ilkhānid courtly circles by Shams al-Din Āmoli (d. 1352), whose *Jewels of Sciences Delightful to Behold* (*Nafayes al-fonun fi ‘ārāyesh al-‘oyun*), an encyclopedic Persian classification of the sciences unprecedented in its polish, scope and depth, served as primary model for subsequent encyclopedists throughout the Persianate world – including Rostamdāri. Most significantly, therein Āmoli posits Sufism (*‘elm-e tasavvof*) as the supreme Islamic science, equal to all the other religious sciences combined, and lettrism as the supreme Sufic science – that is, *the* science of *valāya*. At the same time, he largely adheres to the standard Farabian/Avicennan classification of all occult sciences (including alchemy, letter magic, oneiromancy, physiognomy and astrology) as derivative natural sciences, with one important exception: geomancy, here reclassified as a mathematical science.⁸

By the sixteenth century, then, which featured the entrenchment of millennial cosmocracy as the default form of Islamicate imperialism, these three occult sciences – lettrism, astrology and geomancy – were promulgated among and eagerly seized upon by ruling elites throughout the Persianate world as a philosophically-scientifically rigorous means of harnessing *valāya*. (By contrast, alchemy, while also vigorously pursued by certain scholarly cadres, especially in the western Mamluk-Ottoman context, does not seem to have been de-esotericized and politicized to the same extent.)⁹ Their intellectual cachet and complementarity are testified to by early Safavid philosophers like Shams al-Din Khafri (d. 1535), the leading mathematical astronomer of the century, who naturally authored treatises on both geomancy and lettrism; therein he presents them as sciences that allow the practitioner to explore, and intellectually ascend, the great Chain of Being as proximate cause of all sublunary and celestial phenomena.¹⁰ Likewise, Mir Dāmād (d. 1631), chief founder of the so-called philosophical school of Isfahan and intimate of Shāh ‘Abbās, declared the basis of his metaphysical system as a whole to be emanationist-lettrist Neopythagoreanism: world as (Arabic) text.¹¹

What is rarely acknowledged, however, is the fact that such lofty (occult)philosophizing, this quest for *theosis* (*ta’alloh*), had very immediate political applications. The greatest thinkers of sixteenth- and seventeenth-century Iran, including Mir Dāmād and his philosopher peers, feature in some contemporary sources as practicing occultists in service to the Safavid state, master talismanists responsible for protecting the realm and letter-magically directing its political course. Most prominently among them, Sheykh Bahā’i (d. 1621), the hugely influential Safavid *sheykh al-eslām*, master mathematician and polymath extraordinaire, is remembered to this day as one of the most powerful occultists and letter mages of his generation. But

this function was simply part and parcel of his larger role as a primary architect of the new Safavid imperial culture, in terms material, legal, theological and magical: as an architect, he helped to design and construct Isfahan as imperial capital; as a jurist-theologian, he helped to transform the unstable messianic-Sufi framework of the early Safavid state into a stable orthodox Twelver Shi'i imperial template; as an occultist, he strengthened the state by constructing astral-letter-magical devices – talismans (sg. *telasm*) – to protect and guide the ruling elite and defend against plague and invasion.

Such were the invaluable services charismatic scholar-occultists offered their royal patrons throughout the early modern Persianate world. Most importantly, it was precisely the Neoplatonic/Neopythagorean cosmology these theosized philosophers codified that legitimized Islamicate hierarchical cosmocracy, this lettrist Ebn 'Arabian monism that fueled henological imperialism.

As sketched above, Safavid patronage of occultism was clearly enthusiastic; yet, by comparison to that of their imperial rivals, it appears almost halfhearted. Iranian scholar-occultists were even more in demand at the neighboring Mughal and Ottoman courts, both far wealthier and more cosmopolitan than the Safavid, and many accordingly decamped to seek their fortunes abroad. Indeed, during the sixteenth and seventeenth centuries, the most influential and highest-paid occultists in the Persianate world were typically Shirazi émigrés; thus Shiraz, the city of philosophers and poets, also functioned as the capital of Persianate occultism.¹² The transformation of Iran into a Twelver Shi'i state did little or nothing to impair the prestige of its scholar-occultists in Sunni lands, moreover. The sixteenth century witnessed the careful co-engineering by ruling and scholarly elites of new and distinctive imperial brands across the Persianate world, to be sure, and the imamophilia or confessional ambiguity that had prevailed in preceding centuries – here emblemized by Temür's claim to Alid descent – was indeed replaced by a deliberately hardened confessional division between Shi'i and Sunni.¹³ But the departure this represents is stark only in hindsight, swayed by centuries of imperial and theological propaganda; the Twelverization of Iran, a long and gradual process, should not distract us from the much deeper cultural continuities that made the Persian cosmopolis a cohesive unit until at least the nineteenth century and one of history's vastest.¹⁴ Most importantly, early modern Muslim dynasts, Sunni and Shi'i alike, continued to compete fiercely over the category of *valāya*, still accessible solely through the integrated spiritual-scientific complex of Sufism, occultism and Alidism. Thus, while Safavid Iran was poorer, weaker and more insular than its imperial competitors – to the extent that it is difficult to justify calling it an empire¹⁵ – its intellectual and cultural influence from the Balkans to India was comparatively outsized; and not a little of that influence was due to Iran's continuing status as preeminent producer and exporter of occultists.

This chapter thus examines the occult sciences as theorized and practiced in Safavid Iran during the sixteenth and seventeenth centuries, with attention to both Iranian scholar-occultists who chose to remain under Safavid patronage and the diaspora of those who opted for the greener pastures of the Mughal and Ottoman Empires. For reasons of space, I focus on three occult sciences as illustrative of

developments specific to the Safavid context: geomancy, lettrism and alchemy. This is not to suggest that sciences like astrology and oneiromancy (*'elm al-ta'bir*) were not important – they certainly were, especially at court;¹⁶ but, the very few references in the literature to Safavid occultism tend to fixate on these better-known sciences to the exclusion of their equally important sisters. The account that follows is therefore provisional, and based in the main on unpublished and unstudied sources and artifacts. Nevertheless, the sheer mass of surviving evidence makes one thing clear: we cannot hope to understand the religiopolitical and intellectual history of Safavid Iran, or the full scope of its cultural continuity with its Ottoman, Uzbek and Mughal rivals – and its Renaissance European peers –, without understanding the crucial role of occultism in the construction of a new Safavid imperial culture, and indeed in the transformative *Shi'ization* of Iran.

GEOMANCY

Geomancy, the 'science of sand', is a uniquely Arabic and Islamicate occult science that captured the intellects and imaginations of scholarly elites and their royal patrons throughout the IslamoChristianate world. As a complex divinatory science based on a binary code, geomancy is a younger cognate to the ancient Chinese *I Ching*, but commanded in its prime a far greater territorial spread: it was – and in many cases still is – regularly practiced as a single tradition from Fez, Paris and Timbuktu to Kashgar, Kabul and Delhi, and calved simpler versions throughout sub-Saharan Africa and thence the western hemisphere that remain very much in use. (In Iran in particular, frequent abuse by fraudsters has in fact made its practice into something of a social problem; as a result, *rammali*, 'geomancing', is now legally tantamount to fraud.)¹⁷ Throughout its Afro-Eurasian domain, geomancy was third in popularity only to astrology and oneiromancy; indeed, due to its early incorporation of astrological correspondences, it was often considered a form of 'terrestrial astrology' requiring much less astronomical expertise while remaining richly informative. This association is exemplified by the fact that in the early modern Persianate east individuals designated *monajjems*, or court astrologers, increasingly acted as court geomancers as well.¹⁸ *Raml* and *jafr*, or letter divination, were also often seen as a complementary pair of divinatory techniques based on number theory (*'elm al-'adad*),¹⁹ the first employing dots (*noqāt*) and tetragram figures (*ashkāl*) and the second the letters (*horuf*) of the Arabic alphabet (also derived from a dot), and both equally productive of knowledge of past, present and future.

Together with astrologers and experts in *jafr*, geomancers (sg. *rammāl*) were in high demand at the imperial and regional courts of the Persianate world during the early modern period. Nasir al-Din Tusi (d. 1274), the great Ilkhānid astronomer-philosopher-theologian and director of the Maragha Observatory, is responsible for inaugurating a specifically Persianate geomantic tradition for the benefit of his Mongol patron Hülegü (r. 1256–65); with Tusi as precedent, geomancy went on to exercise some of the best minds of the Persianate world as a mainstream occult-scientific tradition and enjoyed a crescendo of popularity in the sixteenth century with the approach of the Islamic millennium.²⁰ As such, although specialists in

various occult sciences could typically expect to find courtly patronage wherever they traveled, geomancers were among the most mobile and in the sixteenth and seventeenth centuries circulated freely between the Safavid, Mughal and Ottoman realms, while also attracting the patronage of regional courts. Four examples of such individuals must here suffice – three of them, not coincidentally, Shirazis²¹:

- Shāh Tahmāsb, as noted, was obsessed with geomancy. His teacher in the science was one Heydar Rammāl (d. btw. 1559–65), who for eight years was tutor to the young prince before his accession and advisor after.²² Despite his patron's great appreciation for his skill, however, by 1528, Heydar became the target of intrigue by Qezelbāsh amirs seeking to dominate the young shah; the geomancer accordingly fled to the court of Sultan Süleyman (r. 1520–66), where he quickly became an even more influential fixture. Heydar was one of Süleyman's most trusted advisors until at least the 1550s, guiding affairs of state and shaping Ottoman imperial ideology with his regular prognostications and growing immensely wealthy in the process.²³ Geomancy must therefore be considered an important factor in the Ottoman-Safavid conflict from its inception; Sultan Selim (r. 1512–20) had himself relied on court geomancers for military intelligence throughout his astonishing career of conquest – including most notably at the Battle of Chalderan (1514), where Shāh Esmā'il's messianic-magical aura was severely damaged.²⁴
- Shāh Mohammad Khodābanda (r. 1578–87), Tahmāsb's son, would seem to have shared his father's interests: to him was dedicated one of the most important, innovative and comprehensive geomantic manuals of the century, the *World of Geomancy* (*Jahān al-raml*) of 'Abd Allāh Hoseyni Balyāni (fl. 1576), aka Shāh Mollā Monajjem Shirāzi, who first wrote this work for Tahmāsb.²⁵ Most significantly, therein the Safavid *sayyed*-geomancer constructs a conceptual framework for the science that is explicitly 'international': he classifies the regional schools of practice into four broad groups – the Geomantic East (India, Khorasan, Transoxania and points east), the Geomantic North (Fars, Iraq, Gilan), the Geomantic West (the Maghrib, Egypt, Syria, Arabia) and the Geomantic South (the most difficult and occult method, exclusive to the saintly elite in all regions) – and provides a full treatment of each. At the same time, he imparts to it a rich Twelver Shi'i flavor in substantiation of his patrons' total identification of Alidism and occultism.
- Shāh Mollā Monajjem's student – and fellow Shirazi – Hedāyat Allāh Monajjem (fl. 1593) chose rather to seek his fortune in India. To attract Mughal patronage, in 1555 he adapted his teacher's manual to Mughal imperial purposes and dedicated it to Emperor Akbar (r. 1556–1605) in celebration of the latter's impending accession, then rewrote the work in the month of Farvardin in year thirty-seven of Akbar's Elahi calendar – that is, at the spring equinox of the year 1001 (March 1593):²⁶ the moment of dawning of the Islamic millennium over the horizon of Akbar's sacred body, itself effulging forth the talismanic cosmocrator's eschatological imperium over all of India. Entitled *Methods of Guidance* (*Qavā'ed al-hedāya*, a play on the author's name), this revised manual of geomancy was thus one of the many works the Mughal emperor commissioned to celebrate the millennium's arrival, including the feted *Akbarian Statutes* (*A'in-e Akbari*) and *Millennial History* (*Tārikh-e alfi*). Significantly, therein Hedāyat Allāh replaces

the encomium to ‘Ali that figures in the preface to his teacher’s work with a proem heavily astrological in its symbolism, a design feature sure to catch the fancy of Akbar, who had made planetary worship the basis of his daily court ritual – precisely as a means of sidestepping exclusivist Safavid claims to ‘Ali and the *valaya* of the Imams. Despite first being developed for Safavid courtly purposes, its ‘internationalist’ approach to the science was likewise consonant with Akbar’s radically ecumenical doctrine of Universal Harmony (*solh-e koll*); this suggests that a higher degree of continuity obtained between competing Safavid and Mughal imperial ideologies than is usually acknowledged.²⁷

- Emperor Akbar was highly successful in asserting his millennial sovereignty; his rival, Shāh ‘Abbās, was less so. Infamously, near the turn of the millennium the Safavid ruler was warned of an ominous celestial conjunction by his chief court astrologer (*monajjem-bāshi*), who advised him to abdicate the throne for three days in favor of a puppet replacement, then execute his replacement in fulfillment of the celestial sign and reassume the mantle of kingship. For this unhappy role was chosen an adept of the Noqtaviyya movement (a move of some significance, and one to which we will return below).²⁸ The politically expedient nature of this operation aside, it confirms that Shāh ‘Abbās was as reliant on his court occultists for navigating matters of state as his Mughal and Ottoman cognates. The court astrologer in question was Mollā Jalāl al-Din Mohammad Monajjem-e Yazdi (d. 1619), author of *The Abbasid Chronicle* (*Tārikh-e ‘Abbāsi*), covering the years 1576–1611, and *The Astrologers’ Boon* (*Tohfāt al-Monajjemīn*), a comprehensive astrological manual.²⁹ That Mollā Jalāl saw both astrology and geomancy as falling under his professional purview is confirmed by his authorship of the popular manual *Essence of Geomancy* (*Kholāsat al-raml*), written for Khān Ahmad Hoseyni (r. 1536–68, 1577–93), ruler of Gilan.³⁰

Historiographical data thus show Safavid geomancers to be movers and shakers at imperial and regional courts across the early modern Persianate world, responsible for both the crafting of ideology and political and military strategy. This picture is corroborated by the Persian encyclopedic tradition, which progressively canonized the ascent of this science to sociopolitically mainstream status; from the late twelfth century onward, geomancy was reclassified as an applied mathematical science with increasing regularity (as with Rostamdāri above) in order to emphasize its scientific legitimacy, as well as its epistemological continuity with astronomy-astrology.³¹ As Hedāyat Allāh states in the introduction to his *Methods of Guidance*:

Know (God guide you) that philosophy (*hekmat*) is divided into three categories: the natural sciences (*‘elm-e tabi‘i*), the mathematical sciences (*‘elm-e riyyāzi*) and metaphysics (*‘elm-e elāhi*). The principal mathematical sciences are four: astronomy (*hay’at*), arithmetic (*‘adad*), geometry (*handasa*) and music (*musiqi*).³² The mathematical disciplines derived from these are numerous, and include geomancy (*raml*) and astrology (*nojum*), which are exceedingly noble and useful sciences.³³

At the same time, geomancy’s central focus on the four elements – fire, air, water, earth – and the four qualities – hot, cold, wet, dry – also gave it a strong claim to

being a natural science. As the *Tree and Fruit* (*Shajara o samara*), another popular Persian geomantic manual of the period, warns:

The modality of this science is a difficult one, for it is a physiological science ('*elm-e tabāye'*); whoever does not have a firm grasp of the science of natures will not be able to use this science.³⁴

This divinatory art, in other words, occupied the epistemological intersection between astrology, a celestial science, and alchemy, a terrestrial one.

Like all occult sciences as practiced in the early modern Islamo-Christianate world, geomancy is predicated on the hermetic principle of correspondence (*monāsaba*): as above, so below. This union of supernal and infernal, of spirit and matter, of eternity and time is governed by the law of (gendered) attraction or love (*hobb*, *mahabba*, '*eshq*') obtaining between all bodies, from planetary to atomic.³⁵ The Neoplatonists hold that these two principles erect a hierarchy of being, maintained by secondary causation, that may be ascended and descended at will; the Neopythagoreans hold that number is the only shuttle capable of returning the human intellect to its transcendent, unitary origin. Properly manipulated, then, the primordial virtue of number may be activated in a way that grants knowledge of the unseen – past, present and future, essences, hidden objects and secret thoughts. Geomancy is one such method of number manipulation whose efficacy depends on the correspondence of above to below along the Neoplatonic Chain of Being.

This point is explicitly stated by no less a figure than Shams al-Din Mohammad Khafri (resident, naturally, of Shiraz), whom we met above.³⁶ (It is here highly relevant that George Saliba, the historian of science, believes Khafri to be the first thinker to redefine mathematics 'as a descriptive language of scientific processes', foreshadowing 'the modern conception of the role of mathematics in science in general'.)³⁷ In a simply titled *Treatise on Geomancy* (*Resāla dar raml*), one of two Persian works he wrote on the subject – perhaps for Shāh Esmā'il, with whom he was on excellent terms – , Khafri succinctly explains the philosophical-cosmological framework of the science (here with reference to the method Shāh Mollā Monajjem terms the Geomantic East):

Inasmuch as the subject of this science is the dots and the sixteen figures [they form which] populate sixteen cells, it is necessary to explain the nature and associations (*mansubāt*) of each. The methodology of this science is based on correspondence (*monāsabat*); thus because the world of generation and corruption ('*ālam-e kawn o fesād*') is composed of four elements, four single dots are first arranged like so, ∙, ∙, with each standing for a given element. The first dot is called fire, the second air, the third water, the fourth earth. And because the natures (*tabāye'*) of these elements are each composed of two qualities (*keyfiyāt*), each of these four dots may be doubled like so, ∙∙. This figure is known as Populus (*jama'at*), while the first is known as Via (*tariq*). Because Via is the simplest of the figures it is called the Universal Intellect ('*aql-e koll*'), and because Populus features all possible dots it is called the Universal Soul (*nafs-e koll*). Because the eight [composite] qualities of natures are lower than the four elements and the elements lower than the [Universal] Intellect which occupies the second

degree after God Most High, who occupies the first degree of oneness (*yaki*), it is appropriate (*monāseb*) that fire be represented by a single dot, air by two dots, water by four dots and earth by eight dots. This arrangement is also referred to with the letters *ABDH*,³⁸ with *A* assigned to the first dot, *B* to the second, *D* to the third and *H* to the fourth ... And because the cells are sixteen in number, or four times four according to the number of natures, and the sum of the above numbers is fifteen, it is appropriate that the Universal Intellect be assigned to the fifteenth cell, with the sixteenth cell being assigned to the Universal Soul, the closest of contingent beings (*momkenāt*) to the Universal Intellect immediately above it.³⁹

LETRISM

The *mathematization of the cosmos* was, as is well known, the central project driving the so-called Scientific Revolution in Europe (more properly a mathematical revolution), and its culmination with Isaac Newton's (d. 1727) *Principia Mathematica* is still routinely taken to be the basis of European scientific modernity. Yet this project was far less *sui generis* than current scholarship supposes. Evidence suggests that Newton – a committed Neopythagorean-occultist – may well be considered an important member of the *Arabic* astronomical tradition generally and an indirect heir of intellectual developments in Timurid Iran and Transoxania specifically, site of an unprecedented *mathematical turn* in astronomy and a simultaneous *Neopythagorean turn* in metaphysics, with the ascent of lettrism to the status of universal experimental science emblemizing this new quest to mathematize the cosmos.⁴⁰ (It should be borne in mind that Hebrew kabbalah, lettrism's coeval cognate, exploded in popularity among European Renaissance thinkers by the end of the same century and remained a mainstream intellectual pursuit long thereafter.)⁴¹ This quest was first launched by astronomers and lettrists (and astronomer-lettrists) in the employ of Timurid dynasts, who eagerly patronized these men of (occult) science in conscious emulation of their Ilkhānid predecessors (Oloğ Beg's (r. 1409–49) Samarkand Observatory being an upgrade of Hūlegū's (r. 1256–65) at Maragha); communicated to European intellectuals via Jewish intermediaries;⁴² then pursued concurrently but separately in the Persianate and Latinate worlds during the sixteenth and seventeenth centuries.

As heir to this fifteenth-century renaissance of Neopythagoreanism, Khafri, like his contemporary Copernicus (d. 1543), was in the vanguard of those who sought to mathematize the cosmos. It is little surprise, then, that he was also father to the most prolific Persian writer on lettrism of the sixteenth century: Mahmud Dehdār Shirāzi (fl. 1576), teacher of Sheykh Bahā'i in the occult sciences and aspiring courtier to Shāh Esmā'il II (r. 1576–77).⁴³ His son Mohammad, in turn, was an émigré to India and acclaimed author of mystical-philosophical works; he was also the protégé of Mir Fath Allāh Shirāzi (d. 1589), associate of Mir Dāmād, and famously responsible for persuading his 'Adelshāhi patron to lure his teacher to India with a large gift of money, thereby inaugurating a tradition of philosophical instruction in Mughal India under Akbar.⁴⁴ A philosopher, inventor and all-around polymath, Mir Fath Allāh was one of the most prominent Iranian intellectuals of the sixteenth century – and as a Shirazi famed, naturally, for his mastery of the occult sciences.⁴⁵ Mahmud

himself was a student and associate of Sheykh Abo'l-Qāsem Mohammad (d. 1590), aka Amri Shirāzi, who served as court poet under Tahmāsb for thirty years; but Amri, a lettrist and mystical poet with perceived Isma'ili leanings, was blinded by the shah in 1565 on suspicion of associating with the Noqtaviyya, and executed by Shāh 'Abbās as a Noqtavi heretic twenty-six years later.⁴⁶

The souring of Amri's relationship with the Safavid court suggests that a certain tension was developing toward the end of Tahmāsb's reign with respect to the relationship of high lettrism to Noqtavism as the latter became more of a threat to dynastic interests, which may have made Mahmud's official status somewhat tenuous as well. It should be noted here that the Horufiyya movement of Fazl Allāh Astarābādi (d. 1394) and his disciple Mahmud Pasikhāni's (d. 1428) splinter Noqtaviyya movement are typically, but wrongly, assumed in the literature to be representative of fifteenth- and sixteenth-century Persianate lettrism: elite practitioners of the science were consistently adamant as to the theologically perverted, politically dangerous and scientifically worthless nature of these movements' extremist mystical-messianic-materialist approach.⁴⁷ Nevertheless, accusations of Noqtavism were freely hurled by way of intrigue, and often stuck; Mahmud's dedication of two works to Tahmāsb's short-lived and controversial successor, Shāh Esmā'il II, thus, suggests an attempt to reingratiate himself at court. In the preface to the first, entitled *Choicest Secrets and Sayings* (*Zobdat al-asrār va kholāsat al-azkār*) and presumably written in 1576, he significantly provides his royal patron a lettrist proof for the ontological equivalency of Shāh Esmā'il (I and II) and 'Ali – a theme central to early Safavid imperial propaganda – and a prognostication of events up to the Islamic millennium. The second, *Attributes of the Soul: On the Oneness of the All-holy* (*Sefāt al-nofus fi towhid al-qoddus*), opens with versified praise of Mohammad, 'Ali and Shāh Esmā'il II in quick succession.

Possible political difficulties aside, Mahmud Dehdār is responsible for the definitive Twelverization of lettrism, including both letter divination and letter or talismanic magic, such that it became a primary pursuit of Safavid philosophers during the sixteenth and seventeenth centuries. His entire oeuvre – nineteen works survive, all on this science – is explicitly dedicated to this project.⁴⁸ To do so, however, Dehdār did not elide the preceding centuries of intellectual development in favor of a return to its virgin Shi'i source (letter divination, or *jafr*, in particular being an exclusive patrimony of 'Ali b. Abi Tāleb and the members of the Prophetic House, for Shi'i and Sunni alike);⁴⁹ rather, he embraced the state-of-the-art Ebn 'Arabian-Bunian and specifically *Timurid* form of the science, simply nudging it further in the direction of Twelver piety for the benefit of his Safavid patrons. (Significantly, Mir Dāmād too makes Timurid-Aqqoyunlu lettrist theory his primary point of reference.)⁵⁰ He thereby avoided the Shi'i-Sufi extremism of earlier Twelver Shi'i lettrists like the Iraqi scholar Hāfez Rajab al-Borsi (d. 1411), who in his Arabic *Dawning of the Lights of Certain Knowledge: On the True Secrets of the Commander of the Faithful* (*Mashāreq anvār al-yaqīn fi haqā'eq asrār amīr al-mo'menin*) posits the twelve Imams as eternal principles and asserts the divinity of 'Ali on a lettrist basis.⁵¹ Not surprisingly, after a century of neglect, the *Dawning* became a favorite object of commentary in Safavid Iran precisely due to its 'extremist' theological position. But Mahmud's more sober and operative brand of Twelver lettrism, unlike Borsi's, possessed a scholarly legitimacy that was transconfessional, and bore the

prestigious Shirazi stamp; it was hence of immediate use to imamophilic occultists the Persianate world over, especially in Mughal India. As such, a number of his works are well-attested in the manuscript (and lithograph) record, and a few have enjoyed popularity down to the present. Chief among these is his *Keys to All Locks* (*Mafatih al-maghāleq*), a sizable manual of applied lettrism not dissimilar to the *Sun of Knowledge* (*Shams al-ma'āref*), the ubiquitous grimoire associated with the eleventh-century Ifriqian Sufi-mage Ahmad al-Buni;⁵² it appears to have been prized since the sixteenth century and is still readily procurable from bazaars in Iran, with scans circulating freely online. Nor is it dissimilar to such popular contemporary Latinate grimoires as Heinrich Cornelius Agrippa's (d. 1535) *Three Books of Occult Philosophy* (*De occulta philosophia libri tres*); yet the latter has been studied thoroughly by modern historians, and the former not at all.

Representative of Dehdār's Twelverizing approach is his popular treatise *Choicest Talismans* (*Zobdat al-alwāh*), a Persian prose work treating of magical operations based on the *moqatta'at*⁵³ and other subtle points on the science of *jafr* and numbers, including procedures by which to derive the names of angels and jinn for the purpose of constructing incantations (sg. *'azimat*). One such is translated below, a letter-magical invocation of the Fourteen Infallibles (*chahārdah ma'sum*):

Dear friend, know that when one purposes to perform an invocation based on the luminous letters (*da'vat-e nurāni*)⁵⁴ at the beginning of the month, as described above, one must first satisfy several conditions and only then begin. First, before reciting incantations, on each day of these 14 days one should draw a magic square named for one of these fourteen precious souls and seek help from its revelatory spirit, for they are the keepers of the treasure of divine knowledge and the most percipient knowers of the secrets of the unseen realm, so that its great spirit may elevate your state and protect you from danger.

Then, on the first day, having sought by way of blessing the mediation of the purified spirit and fragrant form of the holy Seal of the prophets and master of the messengers, Mohammad, the Messenger of God, arranges a square named for this holy eminence in accordance with the Moon and offer prayers to the revelatory spirit of that holy eminence and his noble House according to the concordant numbers of that square, repeating this litany of prayers the same number of times throughout the fourteen-day period.

On the second day, make a square according to the name of the holy eminence 'Ali b. Abi Tāleb, the triumphant Lion of God, Commander of the Faithful, Leader of the Godfearing and the Faith. On the third day, arrange one according to the name of the exalted eminence and Best of Women Fātema al-Zahrā (God keep her). (Know that the name of her exalted eminence, the Best of Women, is so written [immediately after 'Ali] because the science that the holy eminence and Commander of the Faithful (God keep him) instituted, [a science] that allows one to predict future events until the end of time, is called the Codex (*moshaf*) of Fatema – neglect not this point, and understand!) The same procedure should be followed for the next eleven days, every day constructing a square according to the name of one of the Imams until arriving at the holy eminence and Lord of Command, the Mahdi.

This lowly wretch has expended such extraordinary efforts in this matter and so derived and constructed the abovementioned squares [according to the names of the Fourteen Infallibles] as to surpass all previous efforts, this being an instance of spiritual inspiration; for observing the appropriate times for such operations tells us that they cannot be properly carried out in the absence of a connection with the eternal [divine] emanation. This wretch therefore believes it to be quite possible that no other authority in this field has done something similar, and if they have, it was certainly not with the same degree of finesse.⁵⁵

Although this point has been systematically elided in the literature, then, it was precisely scholar-occultists of the caliber of Mahmud Dehdār who helped to construct a new Safavid Shi'i imperial culture, harnessing and routinizing Shāh Esmā'il's mes-sianic Sufi charisma by means of the most Islamic, and the most Shi'i, of the occult sciences: lettrism. At the same time, their approach was broadly high Persianate and cosmopolitan, with explicit ties to Timurid-Aqqoyunlu precedent, which helped make possible the outsize intellectual and cultural influence of Safavid Iran on its Sunnizing neighbors despite the artificial barrier of confessionalization. Safavid philosophy, in particular, represents the greatest florescence of Islamic metaphysics in the early modern period, as is widely recognized; less recognized is the fact that Safavid philosophers like Mir Dāmād and others saw letter magic as an intrinsic part of philosophical practice.⁵⁶

Dehdār's greatest student is here a case in point. As noted, the Baalbaki Twelver polymath Bahā al-Din al-'Āmeli, better known as Sheykh Bahā'i, comes off in contemporary and traditional sources as one of the most powerful occultists of the Safavid period: master of the magic square, seasoned spellcaster, seer of the unseen, author of fearful illusions.⁵⁷ (It is highly relevant in this connection that he was also famed as an authority on astronomy and the mathematical sciences.) His reputation as an occultist has remained intact down to the present, moreover. Writing in the later nineteenth century, for instance, Mirzā Mohammad b. Soleymān Tonekāboni (d. 1885) offers account upon account of the sheykh's wonderworking ways: his miracles (*karāmāt*) include conversation with the dead, the creation of dragons and the taming of lions.⁵⁸ His talismans are reported to have protected Shiraz and Isfahan from the plague for centuries:

It is widely known that Sheykh Bahā'i engraved a magic square (*morabba'*) or other figure on a stone and buried it on Iran's frontier in the vicinity of Shiraz to defend against the woe of plague (*vabā*), and the plague was thereby prevented from entering Iran from the time of the sheykh to that of Fath 'Ali Shāh [Qājār (r. 1797–1834)], when Prince Hoseyn 'Ali Mirzā was governor of Shiraz and all the princes had designs on the throne and were therefore preoccupied with amassing money. The English were thus able to buy the stone from the prince-governor of Shiraz for 120,000 *tumans*, who out of his lust for wealth turned a blind eye to Iran and her people, forgetting – Say: O God, Master of the kingdom! (Q 3:26) – that kingship is in God's hand alone. In any event, after the stone was sold the plague came to Iran, and pestilence (*ta'un*) followed; and to this day most years see an outbreak of pestilence. The sheykh is also known to have constructed another magic square and buried it in the vicinity of Isfahan

to defend against pestilence, and from that day to this pestilence has not struck the city. Even during the year of the great pestilence that overtook all the cities of Iran, it did not visit Isfahan.⁵⁹

Apart from such reports and Mahmud Dehdār's oeuvre itself, perhaps our best source for understanding the role of occultism in Safavid elite society is, tellingly, a Timurid manual of magic: the *Qasimiyan Secrets* (*Asrār-e qāsemi*) of Hoseyn Vā'ez Kāshefi (d. 1505), preacher, polymath, Naqshbandi Sufi and famed occultist, treating of the sciences of illusionism (*simiyā*) and terrestrial magic (*rimiyā*). This work was first seized upon by Safavid elites in the early sixteenth century – Kāshefi's son 'Ali Safi (d. 1533) produced a simplified version of the work, *Gift for the Khan* (*Tohfa-ye khāni*), for Dormesh Khān Shāmlū, Qezelbāsh commander and governor of Khorasan –, and its popularity blossomed during the seventeenth century. Testifying to the manual's new cachet, in the early decades of that century, it was anonymously expanded to include briefer sections on the three sciences mentioned by the author in the introduction but deliberately left aside: alchemy (*kimiya*), talismans (*limiya*) and planetary magic (*himiya*).⁶⁰

Most significantly for our purposes here, in at least one version these new sections include a wealth of detail as to the identities and activities of practitioners of letter magic during the reign of Shāh 'Abbās in particular – in effect, a Who's Who of high Safavid occultism. The single most-cited authority is, not surprisingly, Sheykh Bahā'i. The influential Avicennan philosopher and Safavid *sadr* Mir Gheyās al-Din Mansur Dashtaki (d. 1542) likewise features as personal lettrist to Shāh Tahmāsb.⁶¹ A number of other prominent scholars and shaykhs are identified as master letter mages, and great emphasis is placed on their prized and potent support of the various political actors of their day – even Ebn Sinā (Sheykh Bu 'Ali) is cited as a lettrist to be feared.⁶² Their number naturally includes Mahmud Dehdār, one of the first authorities to be mentioned. In his case too his service to the Safavid ruling elite is the salient point:

On seeking an audience with kings: One must inscribe a 4 x 4 square on gold and populate it with the numbers of the holy verse, *God is All-gentle with His servants, providing for whomsoever He will* (Q 42:19) [= 998], combining these with the letters of his own name. He should then insert the resulting number in the square, refraining from speech in the process. Of these two names [i.e., his own and the king's], he should write one at the top of the square and one at the bottom ..., and carry [the talisman] on his person. Thus when he sees the king he will be honored and shown generosity. This square was developed by Mowlānā Mahmud Dehdār Shirāzi, who constructed one for Amir Khan;⁶³ as a result he attained control over all Fars.⁶⁴

But lettrists were not simply props to power – they were also checks on that power. Here Sheykh Bahā'i is presented as protecting hapless souls from royal wrath by exerting occult control over the shah's moods:

[On negating the anger of kings]: If a king becomes angry with someone such that the latter is at risk of execution, he should use this same number to inscribe

this square on gold at an auspicious hour and donate some sweetmeats to the poor; in the same hour the king's wrath will turn to graciousness and clemency. My own departed teacher saved many individuals from execution by virtue of this square, and Shaykh Baha' al-Din Mohammad (may He sanctify his secret!) constructed one for Āqā 'Enāyat [Allāh], who as a result was protected from being the object of royal displeasure as long as he lived.⁶⁵

Needless to say, the ability to manipulate the mind and emotions of sovereigns was a two-edged sword. On the one hand, it made lettrists natural allies to ambitious up-and-comers seeking to unseat their superiors; on the other hand, it suggested them as a dangerous liability to any political players who, having achieved power, were more concerned with maintaining the status quo. What all parties could agree on, however, was the incontrovertible effectiveness of lettrist talismans in the pursuit of power. The first passage below presents Dashtaki as talisman-maker extraordinaire to Shāh Tahmāsb; the second attributes the preternatural success of the Qezelbāsh amirs Morshed Qoli Khān (d. 1632) and Allāh Verdi Khān (d. 1613) to their similar dependency on the talismans prepared by philosophers:

On attaining authority over people: One must engrave a 5 x 5 square on copper at an auspicious hour when Venus is in exaltation, populating it with the number of the holy verse *God acts as He wishes and judges as He desires* [= 958].⁶⁶ (This is indeed a rare operation and productive of wonders.) On the back of that square one should write the names *O Haftaya'il*, *O Hanzaya'il*, *O Mukay'il*!, this under Venus, and the operation will be most effective if Venus is in exaltation. This seal talisman (*mohr*) is one of the better-known operations performed by the esteemed sayyed Mir Gheyās al-Din Shirāzi, who constructed one for Shāh Tahmāsb Hoseyni Musavi Safavi. The bearer of this talisman will find all his peers and indeed all men to be obedient to him and he shall be honored by all. My own teacher constructed this talisman for a certain renowned governor (*navvāb-e 'aliyā-ye 'aliyā*) [i.e., Allāh Verdi Khān], who then became ascendant over all rulers (*khavānin-e mo'azzama*) and men and women alike.⁶⁷

On attaining authority over people and one's peers: One must construct a square on gold on Sunday, populating it with the number 958, beginning from the sixteenth cell. The effectiveness of this operation has been thoroughly verified by experience. At the beginning of the process one should perform a ritual ablution, then take various types of precious stones, preferably rubies, and hold them in your mouth; it is best if the precious stones have a hole bored into them. Upon finishing the operation one should not speak to anyone, and one should inscribe the holy verse *God suffices as a witness: Mohammad is the messenger of God* (Q 48:28–29) on the margin of the square. This operation was famously performed by Mowlānā 'Abd al-Latif Gilāni, who constructed [such a talisman] for Morshed Qoli Khan in Mashhad, who then achieved ascendancy over all the amirs and the Qezelbāsh. When one finishes the operation one should give a sizable amount in charity for the poor. My own departed teacher constructed one for Allāh Verdi Khan after the king had appointed him governor (*dārughā*) of Isfahan, who then wore it on his arm; in short order he gained control over all Fars and attained to the greatest glory imaginable.⁶⁸

ALCHEMY

Safavid philosophy, as we have seen, was heavily Neoplatonic and Neopythagorean in orientation, and hence provided a natural basis for occultist practice; the latter, in turn, served admirably the interests of Safavid ruling elites.⁶⁹ It must be emphasized that this philosophical orientation represented not a break with but rather an extension and expansion of the intellectual trends of the Timurid-Aqqoyunlu period, which featured a quest to mathematize the cosmos – a quest that was to have sweeping ramifications well beyond the Persianate world. In particular, this turn away from Avicennan Aristotelianism and toward Neoplatonism and Neopythagoreanism – exemplified by the transmogrification of Ebn Sinā himself, the second Aristotle, into a Neopythagorean-occultist during the Safavid period – was a conscious return to the project of the tenth-century Brethren of Purity (Ekhvān al-Safā), whose seminal and encyclopedic *Epistles* (*Rasā'el*) boomed in popularity in the Persianate east from the fifteenth century onward, especially by way of their abridged Persian translation, *Epitome of Philosophy* (*Mojmal al-hekma*).⁷⁰ In his *Ladder*, for instance, Kāzeruni gushes:

There is not a single word in the *Epistles* that this wretch has not read repeatedly with the eye of desire and longing ... In some of these epistles it is stated that on most topics their discussion accords with the doctrine of Pythagoras.⁷¹

It was therefore seen to be of the utmost intellectual relevance that in their final epistle the Brethren designate alchemy, the queen of the sciences and ultimate goal of philosophy, together with astrology, magic and talismans magic (*'elm al-sehr va-l-telasmāt*), medicine (*'elm al-tebb*) and the 'science of emancipation' (*'elm al-tajrid*).⁷²

Of all the Greek, Egyptian, Syriac, Sabea, Persian and Indian sciences successfully naturalized in the Islamicate context, alchemy was second only to lettrism in its connection with the Imams, and Ja'far al-Sādeq most notably in his role as mentor to the greatest Muslim alchemist in history: Jāber b. Hayyān.⁷³ Regardless of the fact that Jāber may or may not have existed, and if so certainly did not write most of the works attributed to him, the massive Jaberian corpus defined the course of Arabic alchemy until the modern period. Unlike its Hellenic predecessors, moreover, Jaberian alchemy is thoroughlygoingly lettrist in its approach to the analysis of physical objects, positing its highest Science of Balance (*'elm al-mizān*) to be the Balance of Letters (*mizān al-horuf*).⁷⁴ This approach, of course, well accorded with the renaissance of Neopythagoreanism that swept the post-Mongol Islamicate world, and the Shi'i pedigree of both alchemy and lettrism made them all the more attractive to scholar-occultists seeking to Twelverize Iran. In the Safavid period, as a consequence, alchemy was strongly reasserted to be a *revealed* Ja'farian science in the same measure as *jafr*, and as such a prime node for the recovery of the pure Twelver tradition – hence Kāzeruni's and Rostamdāri's valorization of both as sciences of *valāya*.

That dual alchemical-lettrist proclivities pervaded Safavid intellectual culture may be seen in some of the many philosophical anthologies that were produced in the seventeenth century. One impressive anthology, in particular, epitomizes that culture: Tehran MS Dāneshgāh, Dāneshkada-ye Elāhiyāt, 242B. Containing eighty-seven

texts or text excerpts, its strong Neoplatonic-Neopythagorean flavor is produced by the juxtaposition of ancient philosophers like Ps.-Aristotle (i.e., Plotinus, source of the ubiquitous *Theology* (*Othulujiya*)), Alexander of Aphrodisias and Ps.-Apollonius with Muslim philosophers like al-Kendi, Farābi, Ibn Sinā (including the possibly authentic and overtly lettrist *New Year Treatise* (*R. al-Neyruziyya*)), Sohrevardi, Tusi, Qotb al-Din Shirāzi, Jalāl al-Din Davāni and Khafri, followed by several alchemical treatises from the Jaberian corpus and, naturally, a work attributed to Ja‘far al-Sādeq on *jafr*, the last in the anthology. (Significantly, Jāber is here consistently identified as a *Sufi* disciple of Ja‘far.) That is to say, this compendium constructs an intellectual continuum in which both alchemy and lettrism are considered essential components of philosophical practice, especially in a Shi‘izing-Neoplatonizing context.⁷⁵

Intriguingly, contemporary European scholar-occultists like Heinrich Khunrath (d. 1605), John Dee (d. 1608) and Christian Knorr von Rosenroth (d. 1689) were also great proponents of a similar alchemy-kabbalah pairing,⁷⁶ suggesting that it expresses a larger – and as yet almost wholly unexplored – Islamo-Christianate zeitgeist. While early modern Iranian and European intellectuals adopted similar approaches in their common quest for the *philosophia perennis*, however, they did so to very different ends: the latter to prove the historical supremacy of Christian doctrine and restore Edenic human mastery over nature, the former to access and harness the sacral power of the Imams to create a new millennial Safavid Shi‘i society. As that may be, the reductionist-colonialist double standard that still prevails among early modern intellectual and social historians, whereby Christian (and Jewish) occultism is intrinsically susceptible of rehabilitation but Islamic occultism is not, means that scholarship on post-Jaberian alchemical traditions generally and those of the Persianate world specifically is almost totally lacking.

Such a sorry state of affairs, of course, makes comparative study of this remarkable intellectual parallelism impossible, further reinforcing the impression of an East-West philosophical-scientific divergence in the early modern period.⁷⁷ Surviving sources affirm that the scholarly and ruling elites of Safavid Iran were invested in alchemy to no less degree than their European cognates; but we know virtually nothing of how the science was theorized or practiced there, or the extent to which it remained classically Jaberian or absorbed Indian influence. Current research into later Arabic alchemy as pursued in Mamluk Egypt and the Ottoman Empire is, however, suggestive, and has identified certain broad western trends that likely extended to the Persianate east as well. Most notably, Jaberian alchemy enjoyed a renaissance in fourteenth-century Mamluk Cairo through the efforts of Jaber’s greatest heir, the shadowy Āydemir al-Jeldaki (d. after 1344), who elevated it to the status of universal metaphysical-mathematical-natural science;⁷⁸ and another in the seventeenth-century Ottoman context, this on the basis of the massive ‘Alī Çelebi corpus (like the Jaberian corpus largely the work of anonymous authors), which would continue to be of scholarly relevance there to the early twentieth century.⁷⁹ It was also in the seventeenth century that Paracelsian texts were translated from Latin to Arabic for an Ottoman audience⁸⁰ – a development of some significance, given that Paracelsus (d. 1541) himself was interested in the intersections of alchemy and kabbalah, and his seventeenth-century followers even more so.⁸¹

On the practical front, alchemy and economic theory became increasingly intertwined from the Ayyubid period onward, given the obvious attraction of gold

making for the ambitious sultans of the era of Abbasid senescence; overtly alchemical mirrors for princes even began to be produced.⁸² Financial concerns aside, the identification of alchemy with medicine was even more salient for ruling elites, including in post-Mongol Iran. Indeed, the most influential persophone alchemist-lettrist of the late fourteenth century, Sayyid Hoseyn Akhlāti (d. 1397), a Tabrizi, was invited to Cairo to serve as Sultan Barquq's (r. 1382–99) private physician; there he trained the greatest occult philosophers of the next generation, Timurid and Ottoman alike.⁸³ This example alone suggests that a high degree of continuity obtained within the expanding Jaberian – Jeldakian alchemical tradition as variously pursued in Egypt, Anatolia and Iran, with scholar-occultists traveling freely and frequently throughout the region, and Cairo standing as a hub. In the Safavid context, moreover, we do know that there were significant Sufism-alchemy and Noqtavism-alchemy connections,⁸⁴ and that the Sheykhiyya movement of the Qajar period was deeply involved in alchemy and lettrism both.⁸⁵ (The derivative Babi-Bahā'i movement was more focused on lettrism alone.)⁸⁶ The available evidence suggests, in short, that alchemy was a mainstream pursuit of physicians, pharmacologists and mystics in Safavid Iran, as in the Ottoman Empire to the west, and an important part of the distinctively Safavid antiquarianist – perennialist recovery project.

The three founding members of the philosophical school of Isfahan – Mir Dāmād, Sheykh Bahā'ī and Mir Fendereski (d. 1640) – were at the forefront of this project. While the first two focused their energies on theoretical and applied lettrism, however, Mir Fendereski's interests tended more toward alchemy, and he famously spent many years in India pursuing mastery of this science (together with yoga). He was also intimately involved in the great Sanskrit-Persian translation movement under Dārā Shokuh (d. 1659). But unlike many of his compatriots, Mir Fendereski did return to Iran and resumed his teaching of Avicennan philosophy in Isfahan – and continued to practice alchemy.⁸⁷

A significant percentage of the Esfahani philosopher's modest oeuvre treats of the subject; five of his alchemical works, four in Persian and one in Arabic, survive.⁸⁸ Given the very rudimentary stage of scholarship on Persianate alchemy, however, it is unclear how Mir Fendereski's approach relates to the western Jaberian-Jeldakian tradition on the one hand and the Indian traditions he studied on the other, or whether it does incorporate the latter at all. To provide a taste of his method, I translate below the opening and closing passages of his translation-commentary on the seminal Arabic alchemical poem *Nuggets of Gold* (*Shozur al-zahab*) by the Andalusian alchemist Ebn Arfa' Ra's (d. 1197). (That this poem was also the object of two major commentaries by al-Jeldaki suggests that Mir Fendereski's primary point of reference remained the western tradition; his treatise *On Fermentation* (*Resāla takhmiriyya*) likewise cites both Jaber and Mohammad b. Zakariya Rāzi (d. ca. 925)).⁸⁹ This work discusses the production of gold and jewels, as well as the various tools to be employed and the various elements and materials to be used, listing precisely their relative weights and measures; the author goes into some detail about his own alchemical experiments.⁹⁰ He opens as follows:

Know that the elixir (*eksir*) is something that dissolves and coagulates, makes durable while remaining immutable, renders perfect and complete. The substance in which the form of the elixir subsists is subject to a condition whereby

it is not clear whether it is simple (*basit*) or composite (*morakkab*), by which I mean that a given person might see it to be singular, with form corresponding. For a true simple cannot be attained; the world of engendered things is composite in its entirety, and so we cannot attain a simple except that which is so relative to composite things. Thus we must needs inquire into mineral simples, for they alone possess this type of effect. According to this principle, then, we arrive at sulphur (*kebrit*), which when applied to silver (*fezza*) turns it black and brittle, and likewise arsenic (*zernikh*).

If we proceed [on this premise], we take marcasite (*marqashishat*), tutty (*tutiya*) [and] magnesite (*maghnisa*), and find that they indeed exert an effect, but of a type that always results in degeneration (*fesād*). We therefore leave all these aside and take a metal that begins as copper (*nohās*) and which, when combined with silver, can attain to the degree of goldness (*martaba-ye zahabiyyāt*) provided that two things do not prevent this: a lack of patience and a failure to wholly finish the process. And likewise with mercury (*zebāq*) and copper⁹¹

Yet in closing he testifies to failure, together with continued resolve:

With praise to God that my heart's desire has been attained
and my object achieved through divine aid.

My dear friend, many years have I labored in the manner described in this treatise, but have not yet attained to that precious substance – a fact that, I swear by God's own essence, causes me shame before the powers of the habitable earth. But to reach this station is no easy task, and to comprehend and put into operation such complex mysteries is most difficult. May the true Benefactor grant us sustenance through His munificence and largess.

Like a bottle, the whole world strains its neck
to drink the passionwine that reddens its face.⁹²

CONCLUSION

Occultism was a crucial component of Turko-Mongol Perso-Islamic imperial ideologies, generally and key tool in the creation of a new Safavid Shi'i imperial culture specifically; it also drove, and was driven by, the strong Neoplatonic-Neopythagorean turn that defines Safavid philosophy. The Safavid occult philosophers who wrought the Twelverizing transmutation of Iranian society did so explicitly on the basis of Ilkhānid-Timurid-Aqqoyunlu intellectual precedent, while being simultaneously committed to the recovery of the *prisca theologia*, with heavy emphasis on the Greek, Egyptian, Persian and Indian sages of antiquity, disciples of the prophets of God.⁹³ Most significantly, the Safavid antiquarianist pursuit of the perennial philosophy, which posited the Imams as sole vectors of *valaya* and primordial, prophetic knowledge during the Islamic dispensation, was hardly a specifically Twelver innovation, but rather adopted wholesale from the imamophilic Timurid version of the same: by the fifteenth century, the Imams were the patron saints of perennialist occultism for Sunni and Shi'i alike – and continued to be so long after the rise of imperial confessionalization.

In short, the Safavid takeover of Iran, while certainly cataclysmic in some respects, represented less of a cultural rupture than is often assumed. To the contrary, it was precisely the persistence of a high degree of cultural continuity that allowed Safavid scholar-occultists such a wide range of mobility and influence across the Persian cosmopolis. Given the enduring imperial need to access and wield *valāya* in a reliable fashion, they continued to be in high demand at the rival – and much wealthier and more cosmopolitan – Mughal and Ottoman courts, which were often able to poach Safavid scholar-occultists as primary props in the pursuit of world empire.

This contest between sultan-saints and saint-kings for control of Islamdom reached its highest pitch with the approach of the Islamic millennium. The reign of Shāh ‘Abbās, which significantly bracketed that pivotal moment, governed by the momentous 1583 Saturn-Jupiter conjunction immediately preceding it, is universally identified as the apogee of Safavid power and imperial glory. Despite continued patronage of astrologers, geomancers and other occultists at the Safavid court, however, the shah – unlike his Ottoman and Mughal counterparts, Sultan Süleyman and Emperor Akbar – encountered difficulties in claiming millennial sovereignty. In particular, his brutal, hereticidal response to the Noqtavi challenge and the ensuing Noqtavi exodus to India in the late sixteenth century represent a watershed moment in Safavid religiopolitical history whose ideological consequences were far-reaching. Evidence suggests that the affiliated Azari movement, founded by the neo-Zoroastrian ideologue known pseudonymously as Āzar Keyvān (d. 1618) – yet another Shirazi occultist, and associate of Sheykh Bahā’i, Mir Dāmād and Mir Fendereski –, was then attempting to construct a millennial ideology for Shāh ‘Abbās along Persian absolutist, Islamo-Zoroastrian and astrological lines, involving most notably elaborate rituals of planetary worship as pivotal to the performance of cosmic or solar kingship.⁹⁴ This platform was finally rejected by Shāh ‘Abbās, infamously capricious with respect to religious matters, who rather chose to focus on establishing his personal infallibility and absolutist power as centralizer of Iran; a brilliant and sober manager of the Safavid realm responsible for its greatest cultural and economic florescence, and highly attracted to the Timurid model of divinely mandated kingship as well as to certain elements of Noqtavi doctrine, he nevertheless declined to sacralize his royal persona to a degree that would be competitive with those of his Mughal and Ottoman counterparts. Thus spurned, the Azaris decamped for India, and in Emperor Akbar found a patron far more receptive.

Shāh ‘Abbās’s loss of a large contingent of occultists to India was hardly unprecedented, but rather symptomatic of the brain drain Iran had been suffering over the century since its conquest by Shāh Esmā’il. Despite the Safavid dynasts’ heavy reliance on professional occultists, they could not hope to compete financially with their rivals, and anti-Sunni imperial policy, as well as the instability and unpredictability Qezelbāsh hegemony entailed, motivated many of those scholars who wished to stay to seek more reliable patronage abroad.⁹⁵ Mir Fath Allāh Shirāzi is here a prime example: lured to India by financial enticement alone, he quickly attained to awesome influence as a primary architect of Akbar’s new universalist Mughal imperial culture. And Heydar Rammāl, his fellow occultist, could not have had a more assiduous student than Tahmāsb; yet employment at the Sublime Porte proved more reliable and rewarding. It is thus only natural that Safavid shahs would exhibit some insecurity with respect to their failure to retain prominent scholar-occultists. By

the mid-seventeenth century, this insecurity seems to have somewhat compromised the triune Shi'ism-Sufism-occultism model for Safavid legitimacy constructed by Kāzeruni and other scholar-occultists decades earlier. Emblematic of this new tension is the approach of Qotb al-Din Eshkavari Lāhiji (d. btw. 1677–84), a prominent student of Mir Dāmād and *sheykh al-eslām* of Lahijan, whose antiquarianist-perennialist intellectual history, *Hearts' Beloved* (*Mahbub al-qolub*), still insists on the sacral equivalence of Shi'ism and Sufism.⁹⁶ At the same time, however, and his open embrace of occultism as a venerable family tradition notwithstanding,⁹⁷ Eshkavari includes therein the following tendentious anecdote:

One day [Sheykh Bahā'i] entered the majlis of the great king of kings, promulgator of the true Imami rite and master of Iran, Shāh 'Abbās al-Safavi al-Hoseyni (God give his soul to rest in paradise), upon which the shah addressed him: 'Shaykh, just listen to what the ambassador from the king of Byzantium (*malek al-Rum*) [i.e., the Ottoman sultan] has to say!' The ambassador, who was in attendance, began, 'In our country we have a number of scholars who are expert in the occult sciences (*al-'olum al-ghariba*) and magical operations (*al-a'māl al-'ajiba*),' and proceeded to enumerate some of their feats. He then scoffed, 'You have none among your scholars here in Iran who knows these sciences'.

When the sheykh saw that the ambassador's words had troubled the shah and left him at a loss for words, he reassured him, 'The sciences the ambassador has enumerated are considered inconsequential and worthless among true scholars'. While speaking, the sheykh undid the girdle of his robe, and I watched and marveled at the movement of his hand, as did the shah. Suddenly the sheykh lashed forth with his belt in the ambassador's face, gripping one end in his hand, and it instantly transformed into a great dragon (*tennin*). The ambassador and everyone else at the majlis were terrified and leapt up to flee. But the sheykh pulled on the head of the dragon, whereupon it became a belt again. Said the sheykh, in loyal service to the king, 'Such operations mean little to the intelligent. I myself recently learned this operation from a prestidigitator (*ahad men arbāb al-heyal*) in the square of Isfahan. It is a simple act of prestidigitation, and conjurors (*hovāt*) learn such things to relieve the common folk of their *dirhams* and *dinars*'.

The ambassador was left speechless and withdrew from the royal majlis regretting that he had uttered such nonsense in the presence of kings and scholars.⁹⁸

The fact that Sheykh Bahā'i, of all people, is here recruited to delegitimize the occult sciences is eloquent testimony to Safavid insecurity on this point, embodied by the discomfiture of Shāh 'Abbās when faced with the very real superiority of the Ottomans in the field of occultism – a direct consequence of the brain drain from Safavid Iran.

Yet the accusation put into the mouth of the Ottoman envoy is also patently false, as we have seen: Safavid kings and scholars did heavily invest in the occult sciences as an intrinsic part of philosophical practice and the pursuit of empire. And they did so with a considerable measure of success, moreover, this to the point that the Safavid exclusivist claim to 'Ali – the fountainhead of all occult knowledge in Islam – forced Ottoman and Mughal ideologues to find alternative strategies for

claiming the sacral power of the Imams and the saints. The former, capitalizing on the Ottoman Empire's status as New Byzantium, embraced the traditions of occultist Christian and Jewish apocalypticism endemic to the Mediterranean zone, especially those associated with the Hebrew prophet Daniel, and above all Jesus; the latter, still strongly identifying with Iran and Transoxania despite expulsion by the Safavids, synthesized the Azari doctrine of cosmic kingship with the standard Timurid astrological-lettrist platform.⁹⁹ At the same time, Safavid manuals of occultism, like Shāh Mollā Monajjem's *World of Geomancy*, enjoyed popularity beyond Iran, since they could be easily reframed to suit Mughal or Ottoman ideology through the simple expedient of changing their proems, as Hedāyat Allāh Monajjem did for Akbar in celebration of his patron's millennial, astrocratic kingship.

Despite differences of imperial ideological branding, then, the cultural continuity that obtained across the early modern Persianate world is striking in its robustness – and all the more so when we enlarge our analytical aperture to include the far west of the ecumene: Europe. The Safavid court had regular diplomatic contact with its European peers, to be sure, and European sojourners like Chardin reported on Safavid society. But the extent to which intellectual developments in the Persianate and Latinate worlds ran curiously parallel between the fifteenth-seventeenth centuries has yet to be gauged or explained. Here, not surprisingly, the problem of occultism is of particular salience. Revisionist scholarship over the last few decades has shown occultism to be a crucial component of European intellectual and cultural history; Europeanists now widely recognize it to be fundamental to the humanistic projects of Renaissance philosophers and the heroes of the 'Scientific Revolution', and by extension to the imperial ambitions of the European monarchs who patronized them.¹⁰⁰ In particular, the mathematization of the cosmos is far and away the most celebrated outcome of the kabbalist and/or Neopythagorean proclivities of early modern European thinkers. Yet Persianate intellectuals were just as thoroughgoing in their lettrist Neopythagoreanism, and their royal patrons just as reliant on their occult-scientific services – indeed often more so. For the occultist tenor of early modern Islamicate imperialism attained a resonance unmatched across the early modern world stage; and the quest to mathematize the cosmos was first launched in Iran and Transoxania, not in Europe.

Triumphalist teleologies notwithstanding, that this quest did not result in 'scientific modernity' in the post-Mongol Persian cosmopolis but did in the Latin is hardly a tale of Islamic regress vis-à-vis Christian progress, but simply evidence of different metaphysical and cultural priorities as pursued within the strictures of different sociopolitical structures. In the case of Safavid Iran, the desired result was in fact achieved: an entirely new Twelver Shi'i society, wrought alchemically out of Sunni and Shi'i precedent, simultaneously religiopolitically distinctive and intellectually universal. We must therefore retire the Whiggish doctrine of East-West rupture that is still assumed in most scholarship on early modern intellectual or social history; but to do so, we must first exorcise the occultophobia that haunts Islamic studies. For Safavid Iran was no lonely island in the ocean of early modernity, but occult-scientific capital of the Persian cosmopolis, pivot of the ecumene. Its imperial culture is inexplicable absent the occult philosophers who stayed to construct and maintain it; its cultural reach from India to England is inconceivable absent the diaspora of Safavid occultists who won fame and furthered empire abroad.

NOTES

- 1 Massé 1954: 240.
- 2 This rubric, meaning those sciences that are unusual, rare or difficult – i.e., elite sciences, or sciences of the paranormal –, includes astrology, alchemy and variety of magical and divinatory techniques, routinely designated as such in encyclopedias of the sciences, chronicles, biographical dictionaries, legal and theological tracts, etc.; less frequently used terms are *‘olum khafiya* or *ghamedā*, sciences that are hidden or occult. More generally, ‘science’ awkwardly translates the Arabic term *‘elm* (lit., ‘knowledge’), which rather corresponds to the broader concept of Latin *scientia* or German *Wissenschaft*, even *Geisteswissenschaft*.
- 3 D’Alessandri 1873: 223; Fleischer 2009: 241. On Tahmāsb’s astrological interests see Tourkin 2003.
- 4 Kāzeruni 1386/2007: 81–130.
- 5 This encyclopedia is unpublished; see Melvin-Koushki 2016.
- 6 Moin 2012a: 23–55.
- 7 Melvin-Koushki 2016. This includes the Mughal emperor Shāh Jahān (r. 1628–57) most prominently – his Taj Mahal, a masterpiece of Timurid imperial architecture, being expressly designed as a giant letrist riddle (*mo’amma*), and standing as embodiment of paradise and God’s Throne on earth (Parodi 2000).
- 8 See Melvin-Koushki 2018b.
- 9 My thanks to Tuna Artun for this observation.
- 10 See Melvin-Koushki 2018b.
- 11 Melvin-Koushki 2019.
- 12 See Anooshahr 2014.
- 13 Melvin-Koushki 2012: 69–77. This confessional ambiguity is termed Alid loyalism by Marshall Hodgson and Twelver Sunnism (*tasannon-e esnā-‘ashari/davāzdah emāmi*) by Mohammad Taqī Dāneshpazhuh and Rasul Ja’fariyān.
- 14 For a comparative study of the Akkadian, Greek, Latin, Arabic, Persian, Sanskrit and Chinese cosmopolises, see Beecroft 2015.
- 15 Matthee 2010b.
- 16 On Safavid astrology see Babayan 2002; Brentjes 2010. On Safavid oneiromancy, a favorite means of establishing contact with the Imams, see Ahmadi 2009; Babayan 2009. Tahmāsb was a particular devotee of the latter science; as he avers in his autobiography:

It is the firm belief of this poor slave, Tahmāsb al-Safavi al-Musavi al-Hoseyni, with respect to anyone who sees [‘Ali], Commander of the Faithful, in a dream, that whatever that holy eminence declares must come to pass; in this there is no doubt.

See Tahmāsb 1383/2004: 70–1; various of his dreams are recorded at 58, 71–2, 96, 97, 126, 142, 143.
- 17 Doostdar 2018: 50–1.
- 18 Fahd 1966: 41, 200.
- 19 That is, the ‘science of number’, or arithmetic, particularly as received from Pythagoras and Nicomachus.
- 20 Melvin-Koushki 2018b.
- 21 Grey 1873: 223; Fleischer 2009, 241. On Tahmāsb’s astrological interests see Tourkin 2003.
- 22 He is perhaps to be identified with the early Safavid geomancer Heydar b. Mohammad Rammal Esfahani Tabrizi, author of the popular geomantic manual *Kashf al-asrār*. See Melvin-Koushki 2018b.
- 23 Fleischer 2001; 2009: 240–2.
- 24 Melvin-Koushki and Şen, forthcoming.

- 25 Storey 1977: II.3, 482, no. 851; Melvin-Koushki 2018b. The use of *shāh*, a Persian royal title increasingly adopted by prominent Sufi masters from the fourteenth century onward in reference to the Ebn ‘Arabian doctrine of cosmic rule by saints, here refers to the occultist’s *sayyed* (i.e., Alid) status; as such, it neatly encapsulates the Safavid Alidism-occultism-Sufism complex referred to above.
- 26 Storey 1977: II.3, 482, no. 850.
- 27 Melvin-Koushki 2018b. On the smooth Irani-Hindustani continuities defining early modern Persianate culture more generally see Kia 2020.
- 28 Babayan 2002: 350–1.
- 29 Quinn 2000: 21–2.
- 30 Storey 1977: II, 3, 481, no. 849; Melvin-Koushki 2018b.
- 31 Melvin-Koushki 2016.
- 32 I.e., the *quadrivium*.
- 33 MS Majlis 12563: 312; see also Brentjes 2010.
- 34 MS Majlis 12563: 820.
- 35 I.e., the erotic motion of Plotinus (Rizvi 2009: 85). Most famously the basis for Isaac Newton’s theory of gravitation, this concept was ubiquitous in early modern Islamo – Christian thought.
- 36 On Khafri see Saatchian 2011.
- 37 Saliba 2012.
- 38 I.e., 1–2–4–8.
- 39 MS Majles 15755/2 ff. 37b–38a; see Melvin-Koushki 2018b.
- 40 Melvin-Koushki 2017; on the mathematical turn see Saliba 1994; Ragep 2004; Fazlıoğlu 2014.
- 41 Giovanni Pico della Mirandola (d. 1494), putative father of humanism, is usually identified as the first Christian kabbalist; famously, he posited the science as the most effective means of harmonizing Plato and Aristotle and achieving the divinization of man. See Wirszubski 1989.
- 42 Ragep 2004: 130, 139; Morrison 2014.
- 43 For a full treatment see Melvin-Koushki 2016.
- 44 Golchin-e Ma‘ani 1990: no. 451; Rizvi 2011: 9–11; Anooshahr 2014.
- 45 See e.g., Bada‘uni 1379/2000: 105.
- 46 Daftary 2012: 19–20.
- 47 Melvin-Koushki 2012: 41718; 2016. Mahmud’s son Mohammad, writing in defense of his father, is quite explicit on this point (Mohammad Dehdār 1375/1996: 105 130–1, 141, 319. Such defensive characterizations of these movements as extremist do not make them so, of course – but nor do they make them representative of a much larger historical phenomenon. On Safavid Noqtavism, see Babayan 2002; Amanat 1996.
- 48 Predictably, none have been published to date.
- 49 Relevant in this connection is Safavid elite and popular enthusiasm for the art of bibliomancy (*fāl*), using both the Quran and the *divān* of Hāfez, Shirazi Tongue of the Unseen (*lesān al-ghayb*). See Gruber 2001.
- 50 Melvin-Koushki 2019.
- 51 On this work, see Lawson 1999.
- 52 See Gardiner 2014.
- 53 I.e., the disconnected letters that open certain Qur’anic suras, held by occultists to be reflective of the primordial language of creation and therefore comprising knowledge of past, present and future, and especially political events, to the end of time.
- 54 I.e., the 14 letters of the *moqatta‘āt*. The fourteen remaining letters of the Arabic alphabet are termed ‘dark’ (*zolmāni*).
- 55 Mahmud Dehdār 1883: 17–8.

- 56 Melvin-Koushki 2019.
- 57 Melvin-Koushki 2020; on Sheykh Bahā'i, see Kohlberg.
- 58 Tonekāboni 1383/2004: 291, 293–4; see also al-Mohājer 1373/1994: 229–30, 269–70; 'Abbas 1374/1995: 205–11.
- 59 Tonekāboni 1383/2004: 295.
- 60 *Kimiya*, *limiya*, *himiya*, *simiya* and *rimiya* are often cited in a series, their initial letters being combined to produce the occultist motto 'the whole is a secret' (*KLHSR* = *kolloho serr*). On this work see Subtelny 2020.
- 61 Kāshefi 1883: 85, 92, 97, 104.
- 62 Ibid.: 89.
- 63 Presumably Amir Khān or Amir Beg II Mowsellu, a former Aqqoyunlu commander who joined the Qezelbāsh in 1507 and became one of the most important officers of the Safavid state, holding such posts as guardian of prince Tahmāsb and governor-general of Khorasan. See Woods 1999: 192–3.
- 64 Kāshefi 1883: 85–6.
- 65 Ibid.: 86.
- 66 This 'verse' appears to be a conflation of two separate quranic phrases from Q 14:27 and Q 5:1.
- 67 Kāshefi 1883: 89–90.
- 68 Ibid.: 90.
- 69 On the Neoplatonic – Neopythagorean turn in Safavid philosophy see Rizvi 2007; Pourjavady and Schmidtke 2015; Melvin-Koushki 2019.
- 70 Melvin-Koushki 2017.
- 71 Kazeruni 1386/2007: 205.
- 72 Ekhvān al-Safā' 2011: 95–6. The last science refers to what parapsychologists now term 'out of body experiences' (OBEs).
- 73 De Smet 2008.
- 74 Melvin-Koushki 2012: 181–2.
- 75 Endress 2001: 35–56.
- 76 Forshaw 2013.
- 77 On this problem see Stearns 2011. The most comprehensive and balanced survey of the subject to date, Lawrence Principe's *The Secrets of Alchemy*, still cannot avoid giving the impression that Arabic alchemy faded into irrelevance after its transmission to Europe in the twelfth century.
- 78 Harris 2017; see his forthcoming University of Pennsylvania dissertation, 'Better Religion through Chemistry: The Life and Writings of Aydemir al-Jildakī (fl. 1340s)', the first study on this seminal figure.
- 79 Artun 2013: 114–56, 205.
- 80 Ibid.: 81.
- 81 Forshaw 2013: 376–8.
- 82 My thanks to Nicholas Harris for this and preceding observations.
- 83 Melvin-Koushki 2012: 47–51.
- 84 Lory 1989; Babayan 2002: 73.
- 85 Corbin 2014: 353–5; Cole 1994.
- 86 Melvin-Koushki 2012: 281.
- 87 Rizvi 2005.
- 88 Zādhush 1391/2012: 268–71, 279–80, 311–6, 356–9.
- 89 Ibid.: 269; Artun 2013: 31. It should be noted that *takhmīr*, 'fermentation', can in an alchemical context also mean 'leavening', whereby a mineral or elixiral 'leaven' is introduced into a substance with the object of it functioning in a manner analogous to yeast (my thanks to Nicholas Harris for this observation).

- 90 Zādhush 1391/2012: 271.
- 91 Mir Fendereski 2012: 465–83, 465.
- 92 Ibid.: 483.
- 93 This perennialist bent was a characteristic feature of the illuminationist (*eshraqi*) school of philosophy, dominant in the Persianate world by the sixteenth century; but where Sohrawardi (d. 1191) invoked the ancients as a symbolic means of undercutting certain Avicennan positions, Safavid philosophers actively engaged in collecting and citing pre-Avicennan texts, especially Ps.-Aristotle, Ps.-Farabi and the pre-Socratics (Pourjavady and Schmidtke).
- 94 Moin 2012a; Sheffield 2016.
- 95 Pourjavady 2014: 295.
- 96 This insistence did not entail a defense of organized or popular Sufism: Eshkavari, like his teachers and peers, was contemptuous of such forms, and the concurrent clerical push to delegitimize them as inherently antinomian and/or Sunni succeeded in popularizing anti-Sufism sentiment by the mid-seventeenth century (Anzali 2017).
- 97 Terrier 2011: 187–9.
- 98 Eshkavari 1382/2003: II, 407.
- 99 Melvin-Koushki 2018a.
- 100 For an overview of the burgeoning field of Western esotericism studies see Hanegraaff 2012. John Dee is here the classic example, who in his status as personal occultist to Queen Elizabeth I (r. 1533–1603) famously coined the term *Brytish Impire* to express his millenarian vision of North America as an English-Arthurian patrimony (Dee). On occultism at the Russian court of Peter the Great (r. 1682–1725), see Collis 2012.